

and

$$\bar{y} = \left| \begin{array}{cc} s^2 + 3 & 3 \\ s^2 - 3 & 5 \end{array} \right| \div \left| \begin{array}{cc} s^2 + 3 & -2 \\ s^2 - 3 & s^2 + 5 \end{array} \right| = \frac{2s^2 + 24}{(s^2 + 1)(s^2 + 9)} = \frac{11}{4} \cdot \frac{1}{s^2 + 1} + \frac{3}{4} \cdot \frac{1}{s^2 + 9}.$$

On inversion, we get $x = \frac{11}{4} \sin t + \frac{1}{12} \sin 3t$; $y = \frac{11}{4} \sin t - \frac{1}{4} \sin 3t$

which are the same as the solution in (vii) on p. 499.

Obs. The student should compare the earlier solutions of the above examples with those given now and appreciate the superiority of the transform method over others.

PROBLEMS 21.7

Solve the following simultaneous equations (by using Laplace transforms):

1. $\frac{dx}{dt} - y = e^t$, $\frac{dy}{dt} + x = \sin t$, given $x(0) = 1$, $y(0) = 0$. (U.P.T.U., 2006 ; Delhi, 2002)
2. $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that $x = 2$ and $y = 0$ when $t = 0$. (Kerala, 2005 ; U.P.T.U., 2004)
3. $\frac{d^2x}{dt^2} - x = y$, $\frac{d^2y}{dt^2} + y = -x$, given that at $t = 0$; $x = 2$, $y = -1$, $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$. (P.T.U., 2009 S)
4. $3\frac{dx}{dt} + \frac{dy}{dt} + 2x = 1$, $\frac{dx}{dt} + 4\frac{dy}{dt} + 3y = 0$; given $x = 0$, $y = 0$ when $t = 0$. (Madras, 2003 S)
5. $(D - 2)x - (D + 1)y = 6e^{3t}$; $(2D - 3)x + (D - 3)y = 6e^{3t}$ given $x = 3$, $y = 0$ when $t = 0$.
6. The currents i_1 and i_2 in mesh are given by the differential equations; $di_1/dt - \omega i_2 = a \cos pt$, $di_2/dt + \omega i_1 = a \sin pt$. Find the currents i_1 and i_2 by Laplace transform, if $i_1 = i_2 = 0$ at $t = 0$.

21.17 (1) UNIT STEP FUNCTION

At times, we come across such fractions of which the inverse transform cannot be determined from the formulae so far derived. In order to cover such cases, we introduce the *unit step function* (or *Heaviside's unit function**).

Def. The unit step function $u(t - a)$ is defined as follows:

$$u(t - a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t \geq a \end{cases}$$

where, a is always positive (Fig. 21.3). It is also denoted as $H(t - a)$.

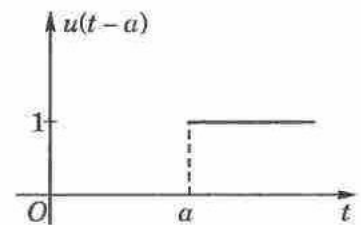


Fig. 21.3

(2) Transform of unit function.

$$L\{u(t - a)\} = \int_0^\infty e^{-st} u(t - a) dt = \int_0^a e^{-st} \cdot 0 dt + \int_a^\infty e^{-st} \cdot 1 dt = 0 + \left[\frac{e^{-st}}{-s} \right]_a^\infty$$

Thus $L\{u(t - a)\} = e^{-as}/s$.

The product $f(t) u(t - a) = \begin{cases} 0 & \text{for } t < a \\ f(t) & \text{for } t \geq a. \end{cases}$

The function $f(t - a) \cdot u(t - a)$ represents the graph of $f(t)$ shifted through a distance a to the right and is of special importance.

Second shifting property. If $L\{f(t)\} = \bar{f}(s)$, then

$$L\{f(t - a) \cdot u(t - a)\} = e^{-as} \bar{f}(s)$$

$$L\{f(t - a) \cdot u(t - a)\} = \int_0^\infty e^{-st} f(t - a) u(t - a) dt$$

*Named after the British Electrical Engineer Oliver Heaviside (1850–1925).

$$\begin{aligned}
 &= \int_0^a e^{-st} f(t-a)(0) dt + \int_a^\infty e^{-st} f(t-a) dt \quad [\text{Put } t-a=u] \\
 &= \int_0^\infty e^{-s(u+a)} f(u) du = e^{-sa} \int_0^\infty e^{-su} f(u) du = e^{-as} \bar{f}(s).
 \end{aligned}$$

Example 21.38. Express the following function (Fig. 21.4) in terms of unit step function and find its Laplace transform. (U.P.T.U., 2002)

Solution. We have $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t-1, & 1 < t < 2 \\ 1, & t > 2 \end{cases}$

$$\begin{aligned}
 f(t) &= (t-1)[u(t-1) - u(t-2)] + u(t-2) \\
 &= (t-1)u(t-1) - (t-2)u(t-2)
 \end{aligned}$$

By second shifting property,

$$L[f(t-a)u(t-a)] = e^{-as} L[f(t)].$$

Also $L[f(t)] = L(t) = 1/s^2$.

$\therefore L[(t-1)u(t-1)]$

$$= e^{-s} \cdot \frac{1}{s^2} \text{ and } L[(t-2)u(t-2)] = e^{-2s} \cdot \frac{1}{s^2}$$

Hence $L[f(t)] = L[(t-1)u(t-1) - (t-2)u(t-2)] = \frac{e^{-s} - e^{-2s}}{s^2}$.

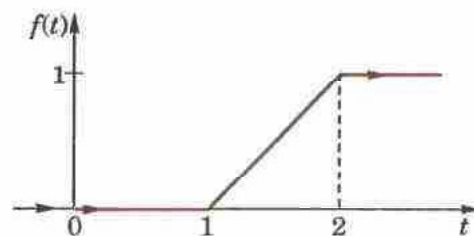


Fig. 21.4

Example 21.39. Using unit step function, find the Laplace transform of

$$f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t < 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases} \quad (\text{V.T.U., 2004})$$

Solution. $f(t) = \sin t [u(t-0) - u(t-\pi)] + \sin 2t [u(t-\pi) - u(t-2\pi)] + \sin 3t \cdot u(t-2\pi)$
 $= \sin t + (\sin 2t - \sin t)u(t-\pi) + (\sin 3t - \sin 2t)u(t-2\pi)$

Since $L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$ and $L(\sin at) = \frac{a}{s^2 + a^2}$,

$$\begin{aligned}
 L[f(t)] &= L(\sin t) + L[(\sin 2t - \sin t) \cdot u(t-\pi)] + L[(\sin 3t - \sin 2t) \cdot u(t-2\pi)] \\
 &= \frac{1}{s^2 + 1} + e^{-\pi s} \left(\frac{2}{s^2 + 4} - \frac{1}{s^2 + 1} \right) + e^{-2\pi s} \left(\frac{3}{s^2 + 9} - \frac{2}{s^2 + 4} \right).
 \end{aligned}$$

Example 21.40. (i) Express the function (Fig. 21.5) in terms of unit step function and find its Laplace transform. (P.T.U., 2005 S)

(ii) Obtain the Laplace transform of $e^{-t}[1 - u(t-2)]$.

Solution. (i) We have $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3. \end{cases}$

$$\begin{aligned}
 f(t) &= (t-1)\{u(t-1) - u(t-2)\} + (3-t)\{u(t-2) - u(t-3)\} \\
 &= (t-1)u(t-1) - 2(t-2)u(t-2) + (t-3)u(t-3)
 \end{aligned}$$

Since $L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$... (λ)

$\therefore L[f(t)] = e^{-s} \cdot \frac{1}{s^2} - 2e^{-2s} \cdot \frac{1}{s^2} + e^{-3s} \cdot \frac{1}{s^2} = \frac{e^{-s}(1 - e^{-s})^2}{s^2}$ [$\because f(t) = t$]

(ii) $L\{e^{-t}[1 - u(t-2)]\} = L(e^{-t}) - L\{e^{-t}u(t-2)\} = \frac{1}{s+1} - e^{-2} L\{e^{-(t-2)}u(t-2)\}$

Taking $f(t) = e^{-t}$, $\bar{f}(s) = \frac{1}{s+1}$ and using (λ) above,

$$L\{e^{-(t-2)}u(t-2)\} = e^{-2s} \cdot \frac{1}{s+1}$$

Hence $Le^{-t}\{1-u(t-2)\} = \{1-e^{-2(s+1)}\}/(s+1)$.

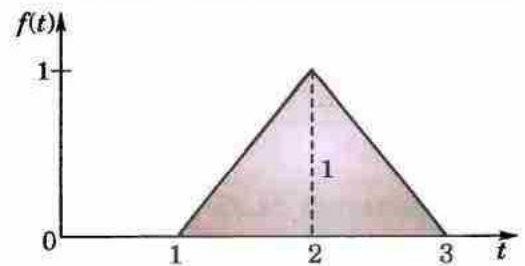


Fig. 21.4

Example 21.41. Using Laplace transform, evaluate $\int_0^\infty e^{-t}(1+2t-t^2+t^3)H(t-1)dt$.

(Mumbai, 2007)

Solution. We have $L\{(1+2t-t^2+t^3)H(t-1)\}$
 $= e^{-s}L[1+2(t+1)-(t+1)^2+(t+1)^3] = e^{-s}L(3+3t+2t^2+t^3)$
 $= e^{-s}\left(3 \cdot \frac{1}{s} + 3 \cdot \frac{1}{s^2} + 2 \cdot \frac{2!}{s^3} + \frac{3!}{s^4}\right) = e^{-s}\left(\frac{3}{s} + \frac{3}{s^2} + \frac{4}{s^3} + \frac{6}{s^4}\right)$

By definition, this implies that

$$\int_0^\infty e^{-st}(1+2t-t^2+t^3)H(t-1)dt = e^{-s}\left(\frac{3}{s} + \frac{3}{s^2} + \frac{4}{s^3} + \frac{6}{s^4}\right)$$

Taking $s = 1$, we obtain

$$\int_0^\infty e^{-t}(1+2t-t^2+t^3)H(t-1)dt = e^{-1}(3+3+4+6) = 16/e.$$

Example 21.42. Evaluate (i) $L^{-1}\left\{\frac{e^{-s}-3e^{-3s}}{s^2}\right\}$

(U.P.T.U., 2002)

(ii) $L^{-1}\left\{\frac{se^{-as}}{s^2-w^2}\right\}$, $a > 0$.

Solution. $L^{-1}\left\{e^{-s} \cdot \frac{1}{s^2}\right\} = \begin{cases} t-1, & t > 1 \\ 0, & t < 1 \end{cases} = (t-1)u(t-1)$

$$L^{-1}\left\{e^{-3s} \cdot \frac{1}{s^2}\right\} = \begin{cases} t-3, & t > 3 \\ 0, & t < 3 \end{cases} = (t-3)u(t-3)$$

$$\therefore L^{-1}\left\{\frac{e^{-s}-3e^{-3s}}{s^2}\right\} = L^{-1}\left(\frac{e^{-s}}{s^2}\right) - 3L^{-1}\left(\frac{e^{-3s}}{s^2}\right) = (t-1)u(t-1) - 3(t-3)u(t-3)$$

(ii) We know that $L^{-1}\left(\frac{s}{s^2-w^2}\right) = \cosh wt$

$$\therefore L^{-1}\left(\frac{se^{-as}}{s^2-w^2}\right) = \begin{cases} \cosh w(t-a), & t > a \\ 0, & t < a \end{cases}$$

$= \cosh w(t-a)u(t-a)$, by second shifting property.

Example 21.43. Find the inverse Laplace transform of :

(i) $\frac{se^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}$ (V.T.U., 2000) (ii) $\frac{e^{-cs}}{s^2(s+a)}$ ($c > 0$).

(Kurukshetra, 2005)

Solution. (i) Since $L^{-1}\frac{s}{s^2+\pi^2} = \cos \pi t$, $L^{-1}\left(\frac{\pi}{s^2+\pi^2}\right) = \sin \pi t$

and

$$L^{-1} [e^{-as} \bar{f}(s)] = f(t-a) \cdot u(t-a) \quad \dots(\lambda)$$

$$\begin{aligned} \therefore L^{-1} \left\{ \frac{se^{-s/2} + \pi e^{-s}}{s^2 + \pi^2} \right\} &= L^{-1} \left\{ e^{-s/2} \cdot \frac{s}{s^2 + \pi^2} \right\} + L^{-1} \left\{ e^{-s} \cdot \frac{\pi}{s^2 + \pi^2} \right\} \\ &= \cos \pi (t-1/2) \cdot u(t-1/2) + \sin \pi (t-1) \cdot u(t-1) \\ &= \sin \pi t \cdot u(t-1/2) - \sin \pi t \cdot u(t-1) = \{u(t-1/2) - u(t-1)\} \sin \pi t \end{aligned}$$

$$(ii) \quad L^{-1} \left\{ \frac{e^{-cs}}{s^2(s+a)} \right\} = L^{-1} \left\{ e^{-cs} \left(-\frac{1}{a^2} \cdot \frac{1}{s} + \frac{1}{a} \cdot \frac{1}{s^2} + \frac{1}{a^2} \cdot \frac{1}{s+a} \right) \right\}$$

Using (λ) above, we have

$$\begin{aligned} L^{-1} \left\{ \frac{e^{-cs}}{s^2(s+a)} \right\} &= -\frac{1}{a^2} [1 \cdot u(t-c)] + \frac{1}{a} [(t-c) \cdot u(t-c)] + \frac{1}{a^2} \{e^{-a(t-c)} \cdot u(t-c)\} \\ &= \frac{1}{a^2} \{a(t-c) - 1 + e^{-a(t-c)}\} u(t-c). \end{aligned}$$

Example 21.44. A particle of mass m can oscillate about the position of equilibrium under the effect of a restoring force mk^2 times the displacement. It started from rest by a constant force F which acts for time T and then ceases. Find the amplitude of the subsequent oscillation.

Solution. The constant force F acting from $t = 0$ to $t = T$ can be expressed as

$$F [1 - u(t-T)], \quad 0 < t < T$$

$\therefore$ equation of motion of the particle is

$$m \frac{d^2 x}{dt^2} = F [1 - u(t-T)] - mk^2 x \quad \text{or} \quad \frac{d^2 x}{dt^2} + k^2 x = \frac{F}{m} [1 - u(t-T)]$$

Taking Laplace transform of both sides, we get

$$(s^2 + k^2) \bar{x} = \frac{F}{ms} (1 - e^{-sT}) \quad [\because x = 0, \dot{x} = 0 \text{ at } t = 0]$$

or

$$\begin{aligned} \bar{x} &= \frac{F}{m} \cdot \frac{1 - e^{-sT}}{s(s^2 + k^2)} = \frac{F}{m} (1 - e^{-sT}) \cdot \frac{1}{k^2} \left(\frac{1}{s} - \frac{s}{s^2 + k^2} \right) \\ &= \frac{F}{mk^2} \left\{ (1 - e^{-sT}) \frac{1}{s} - (1 - e^{-sT}) \cdot \frac{s}{s^2 + k^2} \right\} \end{aligned}$$

Taking inverse Laplace transform, we obtain

$$x = \frac{F}{mk^2} [(1 - \cos kt) - \{1 - \cos k(t-T)\}] u(t-T)$$

i.e.,

$$x = \frac{F}{mk^2} (1 - \cos kt) \text{ for } 0 < t < T$$

and

$$\begin{aligned} x &= \frac{F}{mk^2} (1 - \cos kt) - \{1 - \cos k(t-T)\} \text{ for } t > T \\ &= \frac{F}{mk^2} \{\cos k(t-T) - \cos kT\} \text{ for } t > T \end{aligned}$$

or

$$x = \frac{2F}{mk^2} \sin \frac{kT}{2} \cdot \sin k(t-T/2) \text{ for } t > T$$

Hence the amplitude of subsequent oscillation (i.e., for $t > T$) = $\frac{2F}{mk^2} \sin \frac{kT}{2}$.

Example 21.45. In an electrical circuit with e.m.f. $E(t)$, resistance R and inductance L , the current i builds up at the rate given by

$$L \frac{di}{dt} + Ri = E(t). \quad \dots(i)$$

If the switch is connected at $t = 0$ and disconnected at $t = a$, find the current i at any instant.

Solution. We have $i = 0$ at $t = 0$ and $E(t) = \begin{cases} E & \text{for } 0 < t < a \\ 0 & \text{for } t > a \end{cases}$

$\therefore$ taking the Laplace transform of both sides, (i) becomes

$$(Ls + R)i = \int_0^{\infty} e^{-st} E(t) dt = \int_0^a e^{-st} E dt = \frac{E}{s} (1 - e^{-as})$$

or

$$i = \frac{E}{s(Ls + R)} - \frac{Ee^{-as}}{s(Ls + R)}$$

On inversion, we get

$$i = L^{-1} \left\{ \frac{E}{s(Ls + R)} \right\} - L^{-1} \left\{ \frac{Ee^{-as}}{s(Ls + R)} \right\} \quad \dots(ii)$$

$$\text{Now } L^{-1} \left\{ \frac{E}{s(Ls + R)} \right\} = \frac{E}{R} \left\{ L^{-1} \left(\frac{1}{s} \right) - L^{-1} \left(\frac{1}{s + R/L} \right) \right\} = \frac{E}{R} (1 - e^{-Rt/L})$$

and

$$L^{-1} \left\{ \frac{Ee^{-as}}{s(Ls + R)} \right\} = \frac{E}{R} \{1 - e^{-R(t-a)/L}\} u(t-a) \quad [\text{By the second shifting property}]$$

Thus (ii) becomes

$$i = \frac{E}{R} \{1 - e^{-Rt/L}\} - \frac{E}{R} \{1 - e^{-R(t-a)/L}\} u(t-a)$$

Hence

$$i = \frac{E}{R} \{1 - e^{-Rt/L}\} \text{ for } 0 < t < a$$

and

$$i = \frac{E}{R} \{[1 - e^{-Rt/L}] - [1 - e^{-R(t-a)/L}]\} = \frac{E}{R} e^{-Rt/L} (e^{-Ra/L} - 1) \text{ for } t > a.$$

Example 21.46. Calculate the maximum deflection of an encastre beam 1 ft. long carrying a uniformly distributed load w lb./ft. on its central half length.

Solution. Taking the origin at the end A, we have

$$EI \frac{d^4 y}{dx^4} = w(x)$$

where $w(x) = w[u(x - l/4) - u(x - 3l/4)]$

Taking the Laplace transform of both sides, (Fig. 21.6), we get

$$\begin{aligned} EI[s^4 \bar{y} - s^3 y(0) - s^2 y'(0) - sy''(0) - y'''(0)] \\ = w \left(\frac{e^{-ls/4}}{s} - \frac{e^{-3ls/4}}{s} \right) \end{aligned}$$

Using the conditions $y(0) = y'(0) = 0$ and taking $y''(0) = c_1$ and $y'''(0) = c_2$, we have

$$EI \bar{y} = w \left(\frac{e^{-ls/4}}{s^5} - \frac{e^{-3ls/4}}{s^5} \right) + \frac{c_1}{s^3} + \frac{c_2}{s^4}$$

$$\text{On inversion, we get } EIy = \frac{w}{24} [(x - l/4)^4 u(x - l/4) - (x - 3l/4)^4 u(x - 3l/4)] + \frac{1}{2} c_1 x^2 + \frac{1}{6} c_2 x^3 \quad \dots(i)$$

$$\text{For } x > 3l/4, \quad EIy = \frac{w}{24} [(x - l/4)^2 - (x - 3l/4)^2] + \frac{1}{2} c_1 x^2 + \frac{1}{6} c_2 x^3$$

and

$$EIy' = \frac{w}{6} [(x - l/4)^3 - (x - 3l/4)^3] + c_1 x + \frac{1}{2} c_2 x^2$$

$$\text{Using the conditions } y(l) = 0 \text{ and } y'(l) = 0, \text{ we get } 0 = \frac{w}{24} \left\{ \left(\frac{3l}{4} \right)^4 - \left(\frac{l}{4} \right)^4 \right\} + \frac{1}{2} c_1 l^2 + \frac{1}{6} c_2 l^3$$

and

$$0 = \frac{w}{6} \left\{ \left(\frac{3l}{4} \right)^3 - \left(\frac{l}{4} \right)^3 \right\} + c_1 l + \frac{1}{2} c_2 l^2$$

whence $c_1 = 11wl^2/192$; $c_2 = -wl/4$.

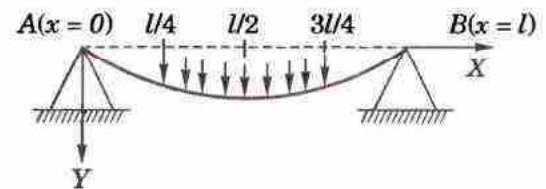


Fig. 21.6

Thus for $l/4 < x < 3l/4$, (i) gives $EIy = \frac{w}{24} \left(x + \frac{1}{4}\right)^4 + \frac{11wl^2}{384} x^2 - \frac{wl}{24} x^3$

Hence the maximum deflection $= y(l/2) = \frac{13wl^4}{6144EI}$.

21.18 (1) UNIT IMPULSE FUNCTION

The idea of a very large force acting for a very short time is of frequent occurrence in mechanics. To deal with such and similar ideas, we introduce the *unit impulse function* (also called *Dirac delta function**).

Thus unit impulse function is considered as the limiting form of the function (Fig. 21.7) :

$$\delta_\epsilon(t-a) = 1/\epsilon, \quad a \leq t \leq a + \epsilon$$

$$= 0, \quad \text{otherwise}$$

as $\epsilon \rightarrow 0$. It is clear from Fig. 21.7 that as $\epsilon \rightarrow 0$, the height of the strip increases indefinitely and the width decreases in such a way that its area is always unity.

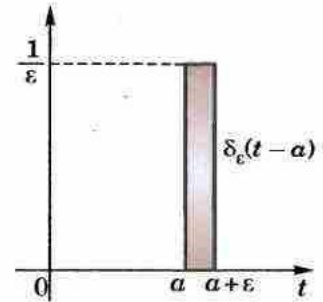


Fig. 21.7

Thus the unit impulse function $\delta(t-a)$ is defined as follows :

$$\delta(t-a) = \infty \text{ for } t = a ; = 0 \text{ for } t \neq a,$$

such that $\int_0^\infty \delta(t-a) dt = 1. \quad (a \geq 0)$

As an illustration, a load w_0 acting at the point $x = a$ of a beam may be considered as the limiting case of uniform loading w_0/ϵ per unit length over the portion of the beam between $x = a$ and $x = a + \epsilon$. Thus

$$w(x) = w_0/\epsilon \quad a < x < a + \epsilon,$$

$$= 0, \quad \text{otherwise}$$

i.e., $w(x) = w_0 \delta(x-a)$.

(2) Transform of unit impulse function. If $f(t)$ be a function of t continuous at $t = a$, then

$$\int_0^\infty f(t) \delta_\epsilon(t-a) dt = \int_0^{a+\epsilon} f(t) \cdot \frac{1}{\epsilon} dt$$

$$= (a + \epsilon - a) f(\eta) \cdot \frac{1}{\epsilon} = f(\eta), \quad \text{where } a < \eta < a + \epsilon.$$

by Mean value theorem for integrals.

As $\epsilon \rightarrow 0$, we get $\int_0^\infty f(t) \delta(t-a) dt = f(a)$.

In particular, when $f(t) = e^{-st}$, we have $L\{\delta(t-a)\} = e^{-as}$.

Example 21.47. Evaluate (i) $\int_0^\infty \sin 2t \delta(t - \pi/4) dt$ (ii) $L\left\{\frac{1}{t} \delta(t-a)\right\}$.

Solution. (i) We know that $\int_0^\infty f(t) \delta(t-a) dt = f(a)$

$$\therefore \int_0^\infty \sin 2t \delta(t - \pi/4) dt = \sin(2 \cdot \pi/4) = 1$$

(ii) We know that $L\{\delta(t-a)\} = e^{-as}$

$$\therefore L\left[\frac{1}{t} \delta(t-a)\right] = \int_s^\infty L\{\delta(t-a)\} ds = \int_s^\infty e^{-as} ds$$

$$= \left[\frac{e^{-as}}{-a} \right]_s^\infty = \frac{1}{a} e^{-as}.$$

* After the English physicist *Paul Dirac* (1902-84) who was awarded the Nobel prize in 1933 for his work in Quantum mechanics.

Example 21.48. An impulsive voltage $E\delta(t)$ is applied to a circuit consisting of L, R, C in series with zero initial conditions. If i be the current at any subsequent time t , find the limit of i as $t \rightarrow 0$?

Solution. The equation of the circuit governing the current i is

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int_0^t i \, dt = E\delta(t) \quad \text{where } i = 0, \text{ when } t = 0.$$

Taking Laplace transform of both sides, we get

$$L[s\bar{i} - i(0)] + R\bar{i} + \frac{1}{C} \frac{1}{s} \bar{i} = E \quad [\text{Using § 21.7 and 21.8}]$$

$$\text{or} \quad \left(s^2 + \frac{R}{L}s + \frac{1}{CL}\right)\bar{i} = \frac{E}{L}s \quad \text{or} \quad (s^2 + 2as + a^2 + b^2)\bar{i} = (E/L)s$$

$$\text{where } R/L = 2a \quad \text{and} \quad 1/CL = a^2 + b^2$$

$$\text{or} \quad \bar{i} = \frac{E}{L} \frac{(s+a) - a}{(s+a)^2 + b^2} = \frac{E}{L} \left\{ \frac{s+a}{(s+a)^2 + b^2} - a \frac{1}{(s+a)^2 + b^2} \right\}$$

On inversion, we get

$$i = \frac{E}{L} \left\{ e^{-at} \cos bt - \frac{a}{b} e^{-at} \sin bt \right\}$$

Taking limits as $t \rightarrow 0, i \rightarrow E/L$

Although the current $i = 0$ initially, yet a large current will develop instantaneously due to impulsive voltage applied at $t = 0$. In fact, we have determined the limit of this current which is E/L .

Example 21.49. A beam is simply supported at its end $x = 0$ and is clamped at the other end $x = l$. It carries a load w at $x = l/4$. Find the resulting deflection at any point.

Solution. The differential equation for deflection is

$$\frac{d^4 y}{dx^4} = \frac{w}{EI} \delta(x - l/4)$$

Taking the Laplace transform, we have $s^4 \bar{y} - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) = \frac{w}{EI} e^{-ls/4}$

Using the conditions $y(0) = 0, y''(0) = 0$ and taking $y'(0) = c_1$ and $y'''(0) = c_2$, we get

$$\bar{y} = \frac{c_1}{s^2} + \frac{c_2}{s^4} + \frac{w}{EI} \frac{e^{-ls/4}}{s^4}.$$

On inversion, it gives $y = c_1 x + c_2 \frac{x^3}{3!} + \frac{w}{EI} \frac{(x - l/4)^3}{3!} u(x - l/4)$

$$\text{i.e., } y = c_1 x + \frac{1}{6} c_2 x^3, \quad 0 < x < l/4 \quad \dots(i)$$

$$\text{and } y = c_1 x + \frac{1}{6} c_2 x^3 + \frac{w}{6EI} (x - l/4)^3, \quad l/4 < x < l$$

Using the conditions $y(l) = 0$ and $y'(l) = 0$, we get

$$0 = c_1 l + \frac{1}{6} c_2 l^3 + 9wl^3/128EI \quad \text{and} \quad 0 = c_1 + \frac{1}{2} c_2 l^2 + 9wl^2/32EI$$

$$\text{whence} \quad c_1 = 9wl^2/256EI, \quad c_2 = -81w/128EI.$$

Substituting the values of c_1 and c_2 in (i), we get the deflection at any point.

PROBLEMS 21.8

1. Represent $f(t) = \sin 2t, 2\pi < t < 4\pi$ and 0 otherwise, in terms of the unit step function and hence find its Laplace transform. (Mumbai, 2005)
2. Sketch the graph of the following functions and express them in terms of unit step function. Hence find their Laplace transforms :

(Assam, 1999)

(i) $f(t) = 2t$ for $0 < t < \pi$, $f(t) = 1$ for $t > \pi$

(ii) $f(t) = t^2$ for $0 < t \leq 2$, $f(t) = 0$ for $t > 2$

(iii) $f(t) = \cos (wt + \phi)$ for $0 < t < T$, $f(t) = 0$ for $t > T$.

3. Express the following functions in terms of unit step function and hence find its Laplace transform.

(i) $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ 1, & \pi < t < 2\pi \\ \sin t, & t > 2\pi \end{cases}$ (V.T.U., 2007)

(ii) $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ \cos 2t, & \pi < t < 2\pi \\ \cos 3t, & t > 2\pi \end{cases}$

(Mumbai, 2008 ; V.T.U., 2003 S)

(iii) $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$

(V.T.U., 2011)

4. Evaluate (i) $L\{e^{t-1}u(t-1)\}$ (ii) $L\{(t-1)^2u(t-1)\}$

(iii) $L(1+2t-3t^2+4t^3)H(t-2)$ (Mumbai, 2007) (iv) $L\{t^2u(t-1)+\delta(t-1)\}$.

5. Evaluate $\int_0^\infty e^{-t}(1+3t+t^2)u(t-2)dt$.

6. Find the inverse Laplace transforms of:

(i) $\frac{e^{-\pi s}}{s^2+1}$

(ii) $\frac{e^{-2s}}{s^2+8s+25}$

(Mumbai, 2006)

(iii) $\frac{e^{-s}}{(s+1)^3}$ (P.T.U., 2010)

(iv) $\frac{3}{s} - 4\frac{e^{-s}}{s^2} + 4\frac{e^{-3s}}{s^2}$

(P.T.U., 2002 S)

7. Solve using Laplace transforms $\frac{d^2y}{dt^2} + 4y = f(t)$ with conditions

$$y(0) = 0, y'(0) = 1 \text{ and } f(t) = \begin{cases} 1 & \text{when } 0 < t < 1 \\ 0 & \text{when } t > 1 \end{cases}$$

(Mumbai, 2007)

8. Using Laplace transforms, solve $x''(t) + x(t) = u(t)$, $x(0) = 1$, $x'(0) = 0$

where $u(t) = \begin{cases} 3, & 0 \leq t \leq 4 \\ 2t-5, & t > 4. \end{cases}$

9. A beam has its ends clamped at $x = 0$ and $x = l$. A concentrated load W acts vertically downwards at the point $x = l/3$. Find the resulting deflection.

$$\left[\text{Hint. The differential equation and the boundary conditions are } \frac{d^4y}{dx^4} = \frac{W}{EI} \delta(x-l/3) \text{ and } \right.$$

$$\left. y(0) = y'(0) = 0, y(l) = y'(l) = 0. \right]$$

10. A cantilever beam is clamped at the end $x = 0$ and is free at the end $x = l$. It carries a uniform load w per unit length from $x = 0$ to $x = l/2$. Calculate the deflection y at any point.

(Kurukshetra, 2006)

[Hint. The differential equation and boundary conditions are

$$\frac{d^4y}{dx^4} = \frac{W(x)}{EI} \quad (0 < x < l) \text{ where } W(x) = \begin{cases} W_0, & 0 < x < l/2 \\ 0, & x > l/2 \end{cases}$$

and $y(0) = y'(0) = 0, y''(0) = y'''(0) = 0]$

11. An impulse I (kg-sec) is applied to a mass m attached to a spring having a spring constant k . The system is damped with damping constant μ . Derive expressions for displacement and velocity of the mass, assuming initial conditions $x(0) = x'(0) = 0$.

$$\left[\text{Hint. The equation of motion is } m \frac{d^2x}{dt^2} = I \delta(x) - kx - \mu \frac{dx}{dt}. \right]$$

21.19 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 21.9

Fill up the blanks or choose the correct answer in each of the following problems :

- Laplace transform of $(t \sin t) = \dots\dots\dots$
- $L\{\delta(t)\} = \dots\dots\dots$
(a) 0 (b) e^{-as} (c) ∞ (d) 1.
- If $L\{f(t)\} = f(s)$, then $L\{e^{-at}f(t)\}$ is
(a) $f(s-a)$ (b) $f(s+a)$ (c) $f(s)$ (d) none of these.
- $L(e^{2t} \sin t) = \dots\dots\dots$
- Inverse Laplace transform of $1/(s^2 + 4s + 13) = \dots\dots\dots$
- Laplace transform of $f'(t) = \dots\dots\dots$
- $L^{-1}\left\{\frac{s}{(2s+3)^2}\right\} = \dots\dots\dots$
- $L(\cosh^2 2t) = \dots\dots\dots$
- $L(e^t) = \dots\dots\dots$
- $L(e^{-t} t^k) = \dots\dots\dots$
- $\int_0^\infty e^{-2t} \cos 3t dt = \dots\dots\dots$
- $L[u(t-a)] = \dots\dots\dots$
- $L^{-1}(\sqrt{t}) = \dots\dots\dots$
- If $L\{F(t)\} = f(s)$, then $L\left\{\frac{d^2 F(t)}{dt^2}\right\} = \dots\dots\dots$
- $L(\cos^3 4t) = \dots\dots\dots$
- $L\left(\frac{\sin at}{t}\right) = \dots\dots\dots$
- $L^{-1}\left\{\frac{1}{(s+3)^5}\right\} = \dots\dots\dots$
- $L \cos(2t+3) = \dots\dots\dots$
- $L^{-1}(1/s^n)$ is possible only when n is
(a) zero (b) -ve integer (c) +ve integer (d) negative rational.
- If $L^{-1}\{\phi(s)\} = f(t)$, the $L^{-1}\{e^{-as}\phi(s)\} = \dots\dots\dots$
- $L\{u(t+2)\} = \dots\dots\dots$
(a) e^{-2s}/s^2 (b) e^{2s} (c) $\frac{e^{2s}}{s}$ (d) $\frac{e^{-2s}}{s}$ (V.T.U., 2011 S)
- $L^{-1}\left\{\frac{s^2 - 3s + 4}{s^3}\right\} = \dots\dots\dots$ (V.T.U., 2010 S)
- If $L\{f(t)\} = \bar{f}(s)$, then $L^{-1}\left\{\frac{\bar{f}(s)}{s}\right\} = \dots\dots\dots$
- If $f(t)$ is a periodic function with period T , then $L\{f(t)\} = \dots\dots\dots$
- If y satisfies $y'' + 3y' + 2y = e^{-t}$ with $y(0) = y'(0) = 0$, then $L[y(t)] = \dots\dots\dots$
- $L\{e^{3t}(2 \cos 5t + 3 \sin 4t)\} = \dots\dots\dots$
- $L(4^t) = \dots\dots\dots$
- $L^{-1}\left\{\frac{1}{\sqrt{s+3}}\right\} = \dots\dots\dots$
- Laplace transform of $\sin 2t \delta(t-2)$ is
(a) $e^{2s} \sin 4$ (b) $e^{-2s} \sin 2$ (c) $e^{-4s} \sin 2$ (d) $e^{-2s} \sin 4$. (V.T.U., 2009 S)
- If $L^{-1}\left\{\frac{s}{(s+1)^2}\right\} = \frac{t \sin t}{2}$ then $L^{-1}\left\{\frac{8s}{(4s^2+1)^2}\right\} = \dots\dots\dots$ (P.T.U., 2009)
- $L^{-1}\{e^{-as} F(s)\} = \dots\dots\dots$
(a) $f(t) u(t)$ (b) $f(t-a) u(t)$ (c) $f(t-a) u(t-a)$ (d) None of these. (V.T.U., 2009 S)
- $L^{-1}\left\{\frac{1}{(s+a)^2}\right\} = \dots\dots\dots$
(a) e^{at} (b) e^{-at} (c) te^{-at} (d) te^{at} (e) $-t$.

34. Laplace transform of $t^4 e^{-at}$ is

(i) $\frac{4!}{(s+a)^4}$

(ii) $\frac{4!}{(s+a)^5}$

(iii) $\frac{4!}{(s-a)^4}$

(iv) $\frac{5!}{(s-a)^5}$

35. Laplace transform of $te^{at} \sin(at)$, $t > 0$, is

(i) $\frac{s-a}{(s-a)^2+a^2}$

(ii) $\frac{a(s-a)}{(s-a)^2+a^2}$

(iii) $\frac{2a(s-a)}{[(s-a)^2+a^2]^2}$

(iv) $\frac{(s-a)^2}{(s-a)^2+a^2}$

36. $L^{-1} \frac{s^2}{(s^2+4)^2}$ is

(i) $\frac{1}{4} \sin 2t + t \cos 2t$

(ii) $\frac{1}{4} \sin 2t + \frac{1}{2} \cos 2t$

(iii) $\sin 2t + \frac{t}{2} \cos 2t$

(iv) $\frac{1}{2} \sin 2t + \frac{t}{4} \cos 2t$

37. $L^{-1} \frac{1}{s(s^2+1)}$ is

(i) $1 + \sin t$

(ii) $1 - \sin t$

(iii) $1 + \cos t$

(iv) $1 - \cos t$

38. $L[u(t-a)]$ where $u(t-a)$ is a unit step function, is

(i) $\frac{e^{-as}}{s}$

(ii) $\frac{e^{as}}{s}$

(iii) $\frac{e^{-as}}{s^2}$

(iv) $\frac{e^{as}}{s^2}$

(V.T.U., 2011)

39. For a periodic function of period 2π , $\int_{a+2\pi}^{b+2\pi} f(x) dx = \dots\dots\dots$

(P.T.U., 2009)

40. $L[\delta(t-a)]$ where $\delta(t-a)$ is a unit impulse function, is

(i) e^{as}

(ii) e^{-as}

(iii) e^s

(iv) $e^{-as/s}$

(V.T.U., 2010 S)

41. Laplace transform of $\sin^2 3t$ is

(i) $\frac{3}{s^2+36}$

(ii) $\frac{6}{(s^2+36)}$

(iii) $\frac{18}{s(s^2+36)}$

(iv) $\frac{18}{s^2+36}$

(V.T.U., 2010)

42. $[L(t^2 e^{-3t})] =$

(i) $\frac{1}{(s+3)^3}$

(ii) $\frac{2}{(s+3)^2}$

(iii) $\frac{3}{(s+3)^3}$

(iv) $\frac{2}{(s+3)^3}$

(V.T.U., 2011)

43. $\frac{d^2}{ds^2} [L f(t)] - L(t^2 f(t)) = 0$.

(True or False)

44. Laplace transform of $f(t)$ is defined for +ve and -ve values of t .

(True or False)

45. If $L[f(t)] = \phi(s)$, then $L[t f(t)] = \frac{d}{ds} [\phi(s)]$.

(True or False)